

An Lam

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EDUCATION

Master of Science in Computer Science, <i>University of Texas at Arlington</i>	GPA: 4.00 - Dec 2026
Bachelor of Science in Computer Science, <i>Texas Tech University</i>	GPA: 3.96 - Dec 2022

SKILLS

Languages	JavaScript (ES6+), Python, SQL (PostgreSQL, Oracle), C/C++, Bash, HTML5/CSS3
Backend & APIs	Node.js, Express.js, FastAPI, RESTful APIs, Zod, JWT Authentication, CRUD
Frontend	React, Tailwind CSS, Bootstrap
Data Engineering	PostgreSQL, ETL Pipelines, Data Warehousing, Pandas, NumPy, scikit-learn, TensorFlow, Hadoop, MapReduce
Infrastructure & DevOps	Docker, Docker Compose, Nginx, GitHub Actions, CI/CD, Linux, Cron Scheduling
Cloud & Platforms	AWS, Hetzner Cloud, Vercel, Supabase
AI & LLM Integration	OpenAI API, Prompt Engineering, Semantic Search, Cosine Similarity, ChatGPT, Claude, Gemini

WORK EXPERIENCE

Database Administrator PostgreSQL, Python, Node.js, Selenium <i>Phuong Tran Inc</i>	Jan 2023 — Mar 2025 <i>Mentor, Ohio</i>
- Operated a centralized PostgreSQL data warehouse, collecting POS, inventory, and scraped data via automated ETL pipelines for BI reporting. - Built scalable web scrapers using Python and Selenium to monitor 100+ competitor sites and support dynamic pricing. - Developed a full-stack scheduling system (React/Node.js), reducing technician idle time by 1+ hour daily and increasing revenue by 15%. - Optimized query performance using EXPLAIN ANALYZE and indexing to speed up reporting and analytics. - Maintained high-availability POS systems across multiple locations during peak operations.	
Student Researcher – C/C++, Bash, Linux, RISC-V <i>Texas Tech University</i>	Jan 2022 — Dec 2022 <i>Lubbock, Texas</i>
- Engineered automated benchmarking pipelines using Bash/Makefile for large-scale RISC-V data collection. - Analyzed performance data under high-concurrency workloads to identify and resolve system bottlenecks. - Developed visualization dashboards to interpret runtime trends, accelerating technical reporting and diagnostics. - Standardized experimental workflows in Linux to ensure high data integrity for compiler research benchmarks.	

PROJECTS

Movies Big Data Lab - Live Demo <i>PostgreSQL, Node.js, React, Python, Docker Compose, Nginx, GitHub Actions, Hetzner Cloud</i>	Jan 2025 — Present
- Designed and deployed a full-stack data analytics platform using React, Node.js APIs, and a PostgreSQL backend to benchmark database indexing strategies on large datasets. - Built a PostgreSQL analytical system managing 29M+ records, implementing a normalized schema optimized for large-scale query workloads. - Improved PostgreSQL query performance by 4× on a 29M-row dataset using GIN trigram indexes, query plan analysis, and materialized views, reducing query latency from seconds to milliseconds. - Deployed production infrastructure using Docker Compose, Nginx reverse proxy, and automated CI/CD pipelines with GitHub Actions on Hetzner Cloud.	
Portfolio Website <i>React, Tailwind CSS, Framer Motion, OpenAI API, AWS, Docker, CI/CD</i>	Jan 2025
- Engineered a full-stack portfolio platform using React and Tailwind CSS, featuring responsive design and animated UI interactions with Framer Motion. - Developed a semantic search system using cosine similarity to enable intelligent discovery of projects and documents. - Built an AI recruiter assistant using the OpenAI API that retrieves and summarizes resume and project data to answer recruiter queries. - Optimized API token usage through prompt engineering, reducing OpenAI API costs while maintaining response accuracy. - Containerized and deployed the application on AWS using Docker with automated CI/CD pipelines for continuous delivery.	
Distributed Systems & Data Processing <i>C++, Bash, POSIX Threads, MapReduce, Mutex, Hadoop Architecture</i>	Aug 2025 — Sep 2025
- Developed a C++ concurrency framework using POSIX threads and mutexes to coordinate parallel transactions and prevent race conditions. - Designed a two-stage MapReduce processing engine capable of handling 16M+ records, implementing a custom shuffle-and-sort pipeline inspired by Hadoop's distributed architecture.	